Task 1: Research Workflow Management Systems

A workflow is a structured sequence of tasks that are carried out in a particular order to achieve an outcome. A Workflow Management System manages this process by coordinating tasks, considering their dependencies and directing work to the appropriate people or system [8]. From looking at different workflow systems, it became clear that a workflow management system is more than just a list of tasks. It needs to manage how work is created, assigned, routed, completed and monitored throughout the process.

One of the most important concepts identified during the research was the difference between a workflow definition and a workflow instance. A workflow definition is the reusable blueprint of a process. It describes the tasks, actions, transitions and overall structure of the workflow. A workflow instance is one actual execution of that definition and contains its own runtime information, such as its current state and progress [7]. This distinction is important for TIVIDY because the same workflow definition should be able to create multiple independent instances at the same time.

The research also showed that certain responsibilities appear repeatedly across different workflow systems. These include managing workflow, definitions and instances, creating and assigning work items, routing work between activities, responding to events and dependencies, recording workflow history, monitoring execution and communicating with external systems. Workflow systems can also contain nested subprocesses, where one activity contains several smaller activities, gateways and events [4].

Another important finding was that some parts of the system are expected to change independently. For example, the way work is assigned may depend on roles, workload or other rules. Approval and escalation processes may also differ between organisations. Decisions such as routing, assignment and approval can be kept as configurable rules instead of being hard coded into the workflow engine [1]. This suggests that our system should keep these changing behaviours separate from its core workflow execution.

External systems are another area that needs to remain flexible. Workflow systems can communicate with other systems using connectors, API's and messaging [1]. For TIVIDY, this means that the workflow manager should coordinate with external systems without becoming responsible for the functionality of those systems. The research also showed that workflow definitions may change while existing workflows are still running. AWS Simple Workflow Service handles this using versioning, allowing existing instances to continue with their original definition while new instances use the newer version [5].

Overall, the research helped identify what TIVIDY actually needs to be designed around. TIVIDY should be a generic and reusable workflow manager, rather than a system containing logic for one specific process. It needs to manage workflow execution, work items, routing, participants, state, events, history and external integrations. At the same time, behaviours such as routing strategies, approval rules, notifications and external system interfaces should be flexible enough to change without changing the core workflow engine.

References

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Task 2: Define the Scenario and System

1. Initial Scope of TIVIDY
   TIVIDY is a generic workflow management system that coordinates workflows
   Its main responsibilities are to:
   - Store and manage workflow definitions and versions
   - Create and manage workflow instances
   - Create, assign and track work items
   - Route work using configurable rules
   - Mange participants, approvals, reactions and escalation
   - Handle states, events and dependencies
   - Support nested activities (sub processes)
   - Communicate with external systems
   - Record history and audit information
   - Monitor running workflows

2. System Boundaries
   TIVIDY is responsible for coordinating and tracking work, but not for performing the specialised business work itself.
   For example: TIVIDY can assign a doctor an assessment task and track its completion. It also does not replace the hospital's patient, laboratory, pharmacy or billing systems.

3. Workflow definition vs workflow instance
   Workflow definition: reusable blue print of a process. It contains activities, transitions, routing rules, dependencies, approval and escalation rules, events, subprocesses and its version.
   A workflow instance: represents one actual execution of a definition. It contains runtime information such as its current activity and state, assigned participants, completed or pending work, events, errors and history.
   This means that multiple independent instances can run from the same definition.

4. Parts expected to vary
   The following parts should be able to change independently:
   - Assignment and routing: eg. role based, workload based or rule based.
   - Approval and escalation rules: different processes may require different approval levels.
   - Work item behaviour: validation, priority, notifications and security.
   - External integrations: different systems may require different interfaces.
   - Reporting: organisations may require different workflow statistics and reports.

5. Workflow scenario: Hospital Patient Admissions and Treatment

Organisation/Environment

The scenario takes place in a hospital where different departments and roles must coordinate a patient's admission, assessment, treatment and discharge.

Process

The workflow manages a patient journey:
Patient arrives -> Register patient -> Nurse assessment -> Doctor assessment -> determine treatment -> treatment approval (if required) -> treatment -> monitoring -> discharge

Participants/Roles

- Admission staff: register the patient and capture information
- Nurse: perform the initial assessment and record observations
- Doctor: assess the patient and determine treatment
- Senior Doctor/Specialist: handle escalated cases and higher risk approvals
- Laboratory staff: perform requested tests and return results
- Pharmacist: review and dispense approved medication
- Discharge staff: complete the discharge process

Major Work

1. Register patient
2. Perform initial assessment
3. Doctor assessment
4. Request and review tests
5. Determine treatment
6. Obtain approval if required
7. Perform treatment
8. Monitor patient
9. Assess readiness for discharge
10. Discharge patient

Decisions and Routing

- Emergency: follow emergency route
- Tests required: Send request to laboratory
- Treatment requires approval: Send for approval
- Approved?: Continue treatment otherwise escalate/reassess.
- Ready for discharge?: Discharge otherwise continue monitoring.

Approvals and Dependencies

Some treatments require approval from a doctor or senior doctor.
Activities can also depend on previous activities
Eg. Doctor assessment -> test request -> laboratory test -> result -> doctor -> review
The doctor cannot review the result until the laboratory test has been completed

Failures/Exceptions

- Lab failure: retry or alternative route
- Approval timeout: escalate to a senior doctor
- Treatment rejected: reassess treatment
- Patient condition worsens: trigger an emergency workflow
- External system unavailable: retry, wait, notify or escalate

External Systems

- Patient information system
- Laboratory system
- Pharmacy system
- Notification system

TIVIDY coordinates these interactions but does not replace the external systems.

Expected Outcome

The workflow should result in successful treatment and discharge, continued treatment, escalation or controlled failure/cancellation, while retaining the workflow history.